

Lina Haddad

Reliability & Maintenance Engineering Student

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Summary

Final-year engineering student in reliability and maintenance with two internships in rail and cement plants. I combine classical reliability methods (AMDEC, fault trees, Weibull) with Python data analysis to help maintenance teams move from reactive to predictive work. Looking for a final-year internship (PFE) or a graduate role in reliability or maintenance engineering.

Experience

Reliability Engineering Intern | Medjerda Rail Services, Tunis

Jul 2026 - Sep 2026

- Built a Weibull analysis of 6 years of traction-motor failure records to estimate MTBF and propose preventive replacement intervals.
- Ran an AMDEC on the bogie maintenance process with 5 technicians and ranked 38 failure modes by criticality.
- Developed a Python dashboard (pandas, Plotly) tracking MTBF, MTTR and availability for 24 locomotives.

Maintenance Intern | Kerkouane Cement Works, Enfidha

Jul 2025 - Aug 2025

- Collected vibration data on kiln fans and helped the condition monitoring team flag 3 bearing defects before failure.
- Updated preventive maintenance plans in the GMAO (SAP PM) for 40 rotating equipment items.
- Wrote root cause analysis reports (5 Whys, Ishikawa) for recurring conveyor belt stoppages.

Projects

Remaining useful life prediction | NASA C-MAPSS turbofan dataset

- Trained random forest and XGBoost models in sklearn to predict remaining useful life, reaching an RMSE of 18 cycles on the test set.
- Engineered rolling-window features from 21 sensor channels and tuned the models with cross-validation.

Hydraulic press reliability study | Course project

- Modeled the safety system of a hydraulic press with a fault tree and reliability block diagrams in MATLAB and computed system availability.

Education

Engineering degree, Industrial Engineering (Reliability & Maintenance) | Ecole Polytechnique de Dougga | 2024 - 2027 (expected)

Preparatory classes, Mathematics and Physics | 2022 - 2024

Skills

Reliability: AMDEC/FMECA, fault tree analysis, reliability block diagrams, Weibull analysis, MTBF/MTTR, RCM basics

Maintenance: preventive maintenance, condition monitoring, vibration analysis (basics), TPM, GMAO (SAP PM)

Data: Python (pandas, NumPy, scikit-learn, XGBoost), SQL, Power BI, Excel

Tools: MATLAB, SolidWorks, Git

Languages

Arabic (native), French (fluent), English (B2)

Certifications

Lean Six Sigma Yellow Belt (2025)